

Lab06_FernandaVincenzo

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```
library(RSQLite)
```

Warning: pacote 'RSQLite' foi compilado no R versão 4.5.3

1. Conexão

```
conn <- dbConnect(SQLite(), "datasets/uwmadison.sqlite3")  
conn
```

```
<SQLiteConnection>  
  Path: C:\Users\nanda\Documents\ME315\datasets\uwmadison.sqlite3  
  Extensions: TRUE
```

```
dbListTables(conn)
```

```
[1] "course_offerings"  "courses"           "grade_distributions"  
[4] "instructors"       "rooms"              "schedules"  
[7] "sections"         "subject_memberships" "subjects"  
[10] "teachings"
```

2. Professores que lecionaram disciplinas de Estatística (STAT)

```
sql_professores <- "  
SELECT DISTINCT i.id, i.name  
FROM instructors i  
INNER JOIN teachings t      ON t.instructor_id = i.id  
INNER JOIN sections s      ON s.uuid = t.section_uuid
```

```

INNER JOIN course_offerings co      ON co.uuid = s.course_offering_uuid
INNER JOIN subject_memberships sm    ON sm.course_offering_uuid = co.uuid
INNER JOIN subjects su              ON su.code = sm.subject_code
WHERE su.abbreviation = 'STAT'
ORDER BY i.name
"

sql_quant_profs <- "
SELECT COUNT(DISTINCT i.id) AS n_professores
FROM instructors i
INNER JOIN teachings t          ON t.instructor_id = i.id
INNER JOIN sections s          ON s.uuid = t.section_uuid
INNER JOIN course_offerings co  ON co.uuid = s.course_offering_uuid
INNER JOIN subject_memberships sm ON sm.course_offering_uuid = co.uuid
INNER JOIN subjects su          ON su.code = sm.subject_code
WHERE su.abbreviation = 'STAT'
"

```

Nomes e Ids dos professores:

```

profs <- dbGetQuery(conn, sql_professores)
head(profs, 20)

```

	id	name
1	4787911	ABIGAIL BENZINE
2	3111842	ADIN-CRISTIAN ANDREI
3	3775809	AKICHIKA OZEKI
4	4958796	ALAN HUANG
5	5451706	ALBERTO DEL PIA
6	3661782	ALBRECHT KLEMM
7	5656034	ALEXANDER COVINGTON
8	4680082	ALEXANDER FISH
9	3573775	ALEXANDER KISELEV
10	5162339	ALYSSA DIGILIO
11	129415	ANATOLE BECK
12	3674168	ANDREJ ZLATOS
13	5003565	ANDREW LESLIE
14	4952769	ANNE BRUCKNER
15	5498092	ANQI SHI
16	4646729	ANRU ZHANG
17	2600893	ARNOLD MILLER

```

18 4963378          AUGUST JENSEN
19 5852265          BEHZAD AALIPUR
20 5694563 BEHZAD AALIPUR HAFSHEJANI

```

Quantos são:

```

quant_profs <- dbGetQuery(conn, sql_quant_profs)
head(quant_profs)

```

```

      n_professores
1          379

```

3. Dificuldade

```

pontos <- "(gd.a_count*4.0 + gd.ab_count*3.5 + gd.b_count*3.0 +
           gd.bc_count*2.5 + gd.c_count*2.0 + gd.d_count*1.0 + gd.f_count*0.0)"
alunos <- "(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count +
           gd.c_count + gd.d_count + gd.f_count)"

# Base comum: notas de turmas de STAT, ligadas a professor e disciplina
from_base <- "
FROM grade_distributions gd
INNER JOIN sections s          ON s.course_offering_uuid = gd.course_offering_uuid
                                AND s.number = gd.section_number
INNER JOIN teachings t        ON t.section_uuid = s.uuid
INNER JOIN instructors i       ON i.id = t.instructor_id
INNER JOIN course_offerings co ON co.uuid = gd.course_offering_uuid
INNER JOIN courses c           ON c.uuid = co.course_uuid
INNER JOIN subject_memberships sm ON sm.course_offering_uuid = co.uuid
INNER JOIN subjects su         ON su.code = sm.subject_code
WHERE su.abbreviation = 'STAT'
"

```

Professores

```

sql_gpa_prof <- paste0("
SELECT i.id, i.name AS professor,
       SUM(" , pontos, ") / SUM(" , alunos, ") AS gpa,

```

```

SUM(", alunos, ") AS n_alunos
", from_base, "
GROUP BY i.id, i.name
HAVING SUM(", alunos, ") >= ", 1, "
ORDER BY gpa
")
gpa_prof <- dbGetQuery(conn, sql_gpa_prof)

```

Professores mais difíceis (menor GPA):

```
head(gpa_prof, 5)
```

	id	professor	gpa	n_alunos
1	2601633	JAMES D KUELBS	2.637500	40
2	4680082	ALEXANDER FISH	2.638889	36
3	495087	GARY HOWARD SCHROEDER	2.729032	155
4	4752238	SAMUEL STECHMANN	2.741379	29
5	3460189	FANG FANG	2.771127	142

Professores mais fáceis (maior GPA):

```
tail(gpa_prof, 5)[5:1, ]
```

	id	professor	gpa	n_alunos
207	4426932	YAJUAN SI	4.000000	11
206	4272783	GUANHUA CHEN	4.000000	16
205	3884290	SRIKANTHMADHAVAN ARAVAMUTHAN	4.000000	16
204	2600488	GRACE WAHBA	3.971154	156
203	5099221	ZIFENG ZHAO	3.954545	33

Disciplinas

```

sql_gpa_disc <- paste0("
SELECT c.number AS numero, c.name AS disciplina,
      SUM(", pontos, ") / SUM(", alunos, ") AS gpa,
      SUM(", alunos, ") AS n_alunos
FROM grade_distributions gd
INNER JOIN course_offerings co ON co.uuid = gd.course_offering_uuid

```

```

INNER JOIN courses c          ON c.uuid = co.course_uuid
INNER JOIN subject_memberships sm ON sm.course_offering_uuid = co.uuid
INNER JOIN subjects su        ON su.code = sm.subject_code
WHERE su.abbreviation = 'STAT'
GROUP BY c.uuid, c.number, c.name
HAVING SUM(", alunos, ") >= ", 1, "
ORDER BY gpa
")
gpa_disc <- dbGetQuery(conn, sql_gpa_disc)

```

Disciplinas mais difíceis:

```
head(gpa_disc, 5)
```

	numero	disciplina	gpa
1	431	Introduction to the Theory of Probability	2.886374
2	525	Linear Programming Methods	2.945983
3	309	Introduction to Probability and Mathematical Statistics I	3.026515
4	632	Introduction to Stochastic Processes	3.044066
5	310	Introduction to Probability and Mathematical Statistics II	3.057717
	n_alunos		
1	2231		
2	1083		
3	1188		
4	851		
5	771		

Disciplinas mais fáceis:

```
tail(gpa_disc, 5)[5:1, ]
```

	numero	disciplina	gpa
70	834	Empir Proc&Semiparmtrc Infernc	4.000000
69	628	Data Science Practicum	4.000000
68	811	Sample Survey Theory and Method	4.000000
67	841	Nonparametric Statistics and Machine Learning Methods	4.000000
66	803	Experimental Design I	3.984127
	n_alunos		
70	8		
69	12		

68	54
67	24
66	63

4. Desconexão

```
dbDisconnect(conn)
```

Horário de Compilamento

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-20 22:05:41 -03"
```